Prediction Before Interpretation

Testing the empirical standing of field-like constructs in active matter.

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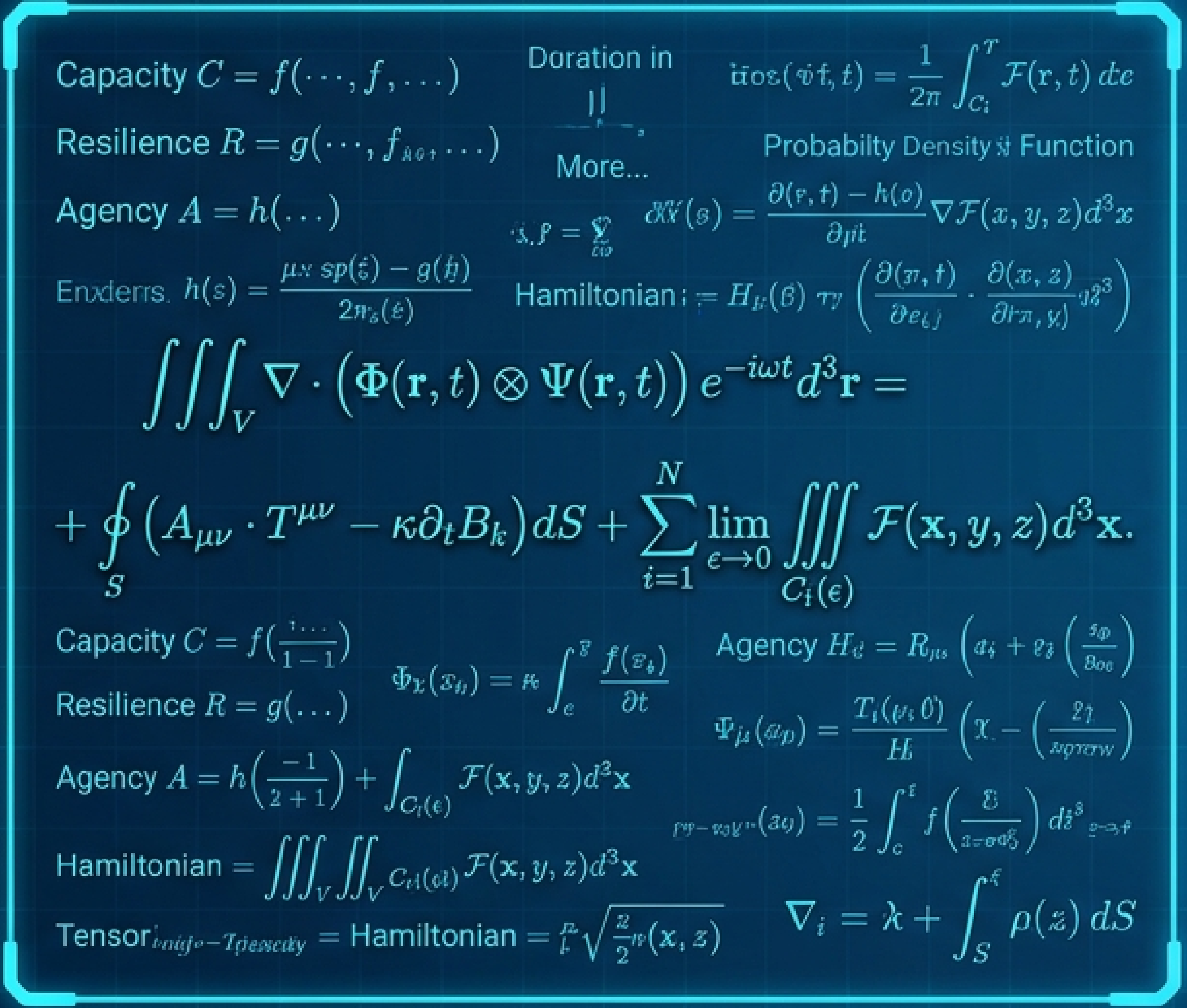
The Pathology of “Field-Like” Constructs

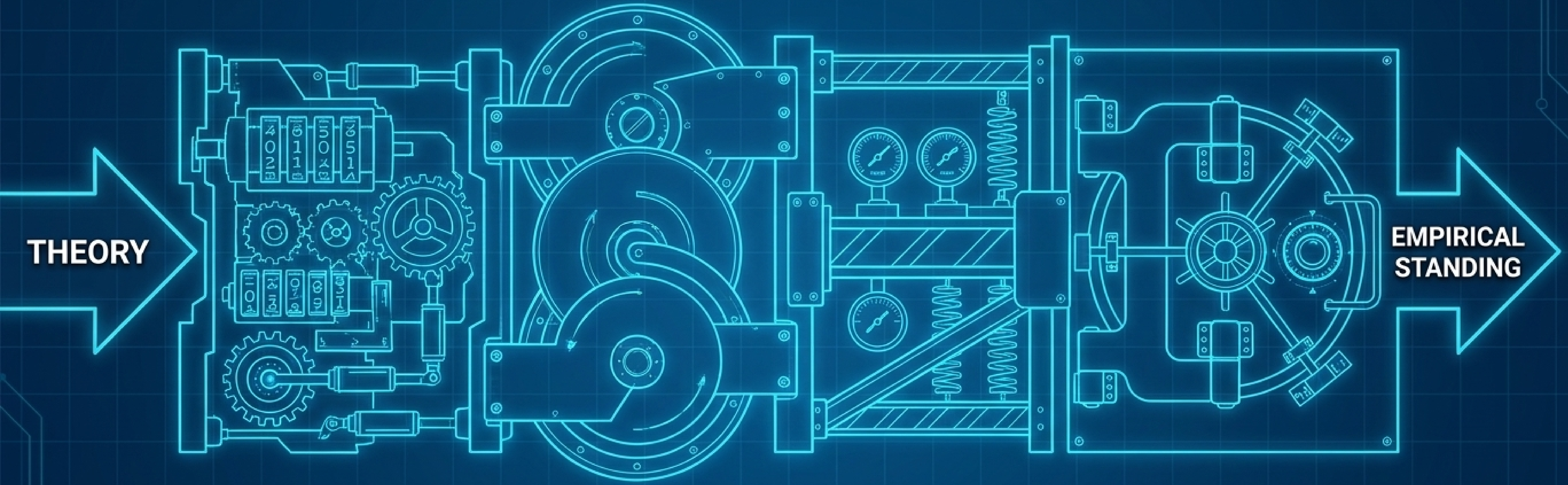
Frameworks frequently introduce terms that behave grammatically like measurable quantities. They are defined with mathematical formality but lack an operational procedure to measure them.



If two competent observers examine the same system and disagree about a construct’s value, is there a procedure that could settle the disagreement?

“New terminology is not new predictive content.”

The 4-Part Identification Criterion



1. Reconstructible

A precise procedure must exist to recover the quantity from raw observational data.

2. Comparable

Competing formalizations of the underlying intuition must be explicitly tested against one another.

3. Stable

Coefficients must survive nuisance parameters (coarse-graining, resolution, sampling).

4. Predictive

The construct must improve out-of-sample prediction beyond the strongest pre-existing baseline.

Pre-Registering the Outcomes

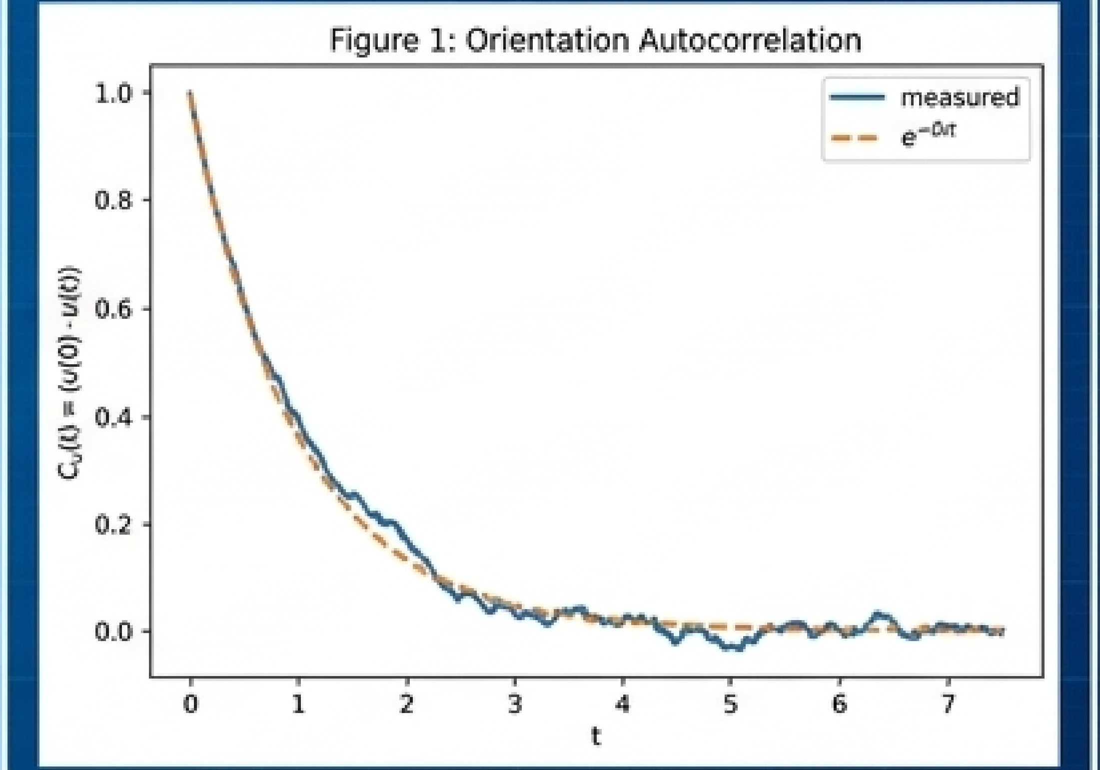
Outcome	Interpretation	Permitted Claim
Strong Success	Predictive improvement holds across sweep + survives intervention	Evidence for mechanistically relevant capacity.
Weak Success	Predictive improvement holds + intervention untested	Early-warning variable, not established as driver.
Conditional Success	Improvement confined to specific regimes	Domain-limited result.
Failure	No predictive improvement beyond baseline	Capacity reconstruction not empirically justified.
Clean Failure	Reconstruction unstable/non-identifiable across scales	Fails before predictive testing.
Indeterminate	Insufficient statistical power	No claim permitted.

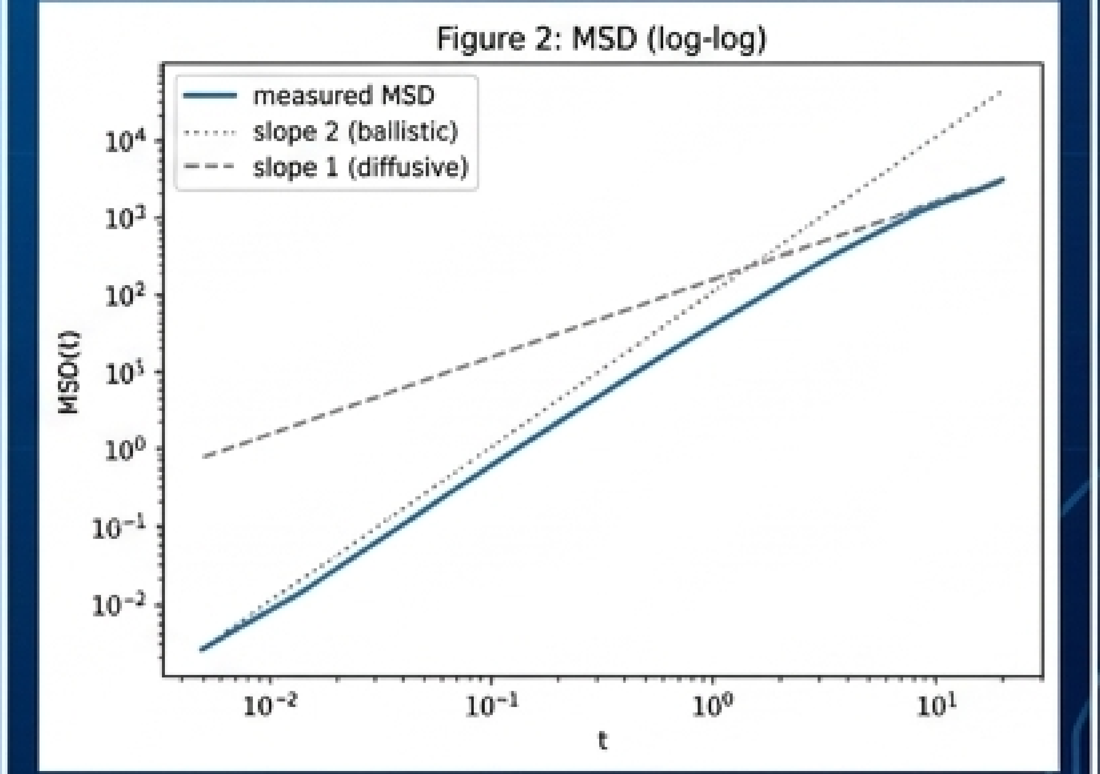
The Test Subject: MIPS in Active Brownian Particles

MIPS was deliberately chosen because it is a hard test. Existing kinetic and hydrodynamic theories already account for the phenomenon without RSVP-style capacity vocabulary. There is no explanatory vacuum to fill.

If a new construct adds predictive power here, the addition is genuinely real. If it adds nothing, the null result is deeply informative.

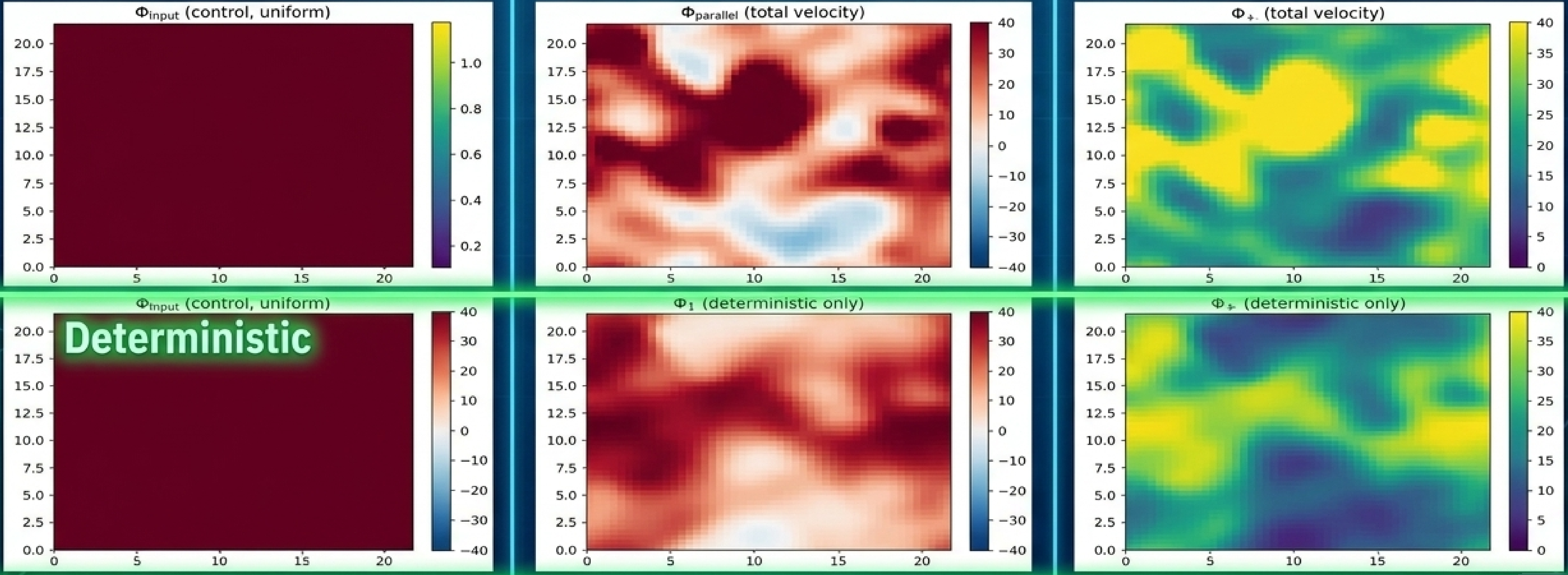
Validation Callout: The engine was built strictly theory-neutral. It knows nothing of “capacity” and emits only raw trajectories and forces.





The Competitors: Three Operationalizations of ‘Capacity’

Φ_{input} (Imposed)	$\Phi_{parallel}$ (Signed Realized)	Φ_{+} (Usable Forward)
The imposed propulsion speed. Trivially uniform; retained as a null control.	Alignment between propulsion direction and actual velocity. Can go negative during collisions.	The non-negative part of realized alignment. Discards backwards motion.



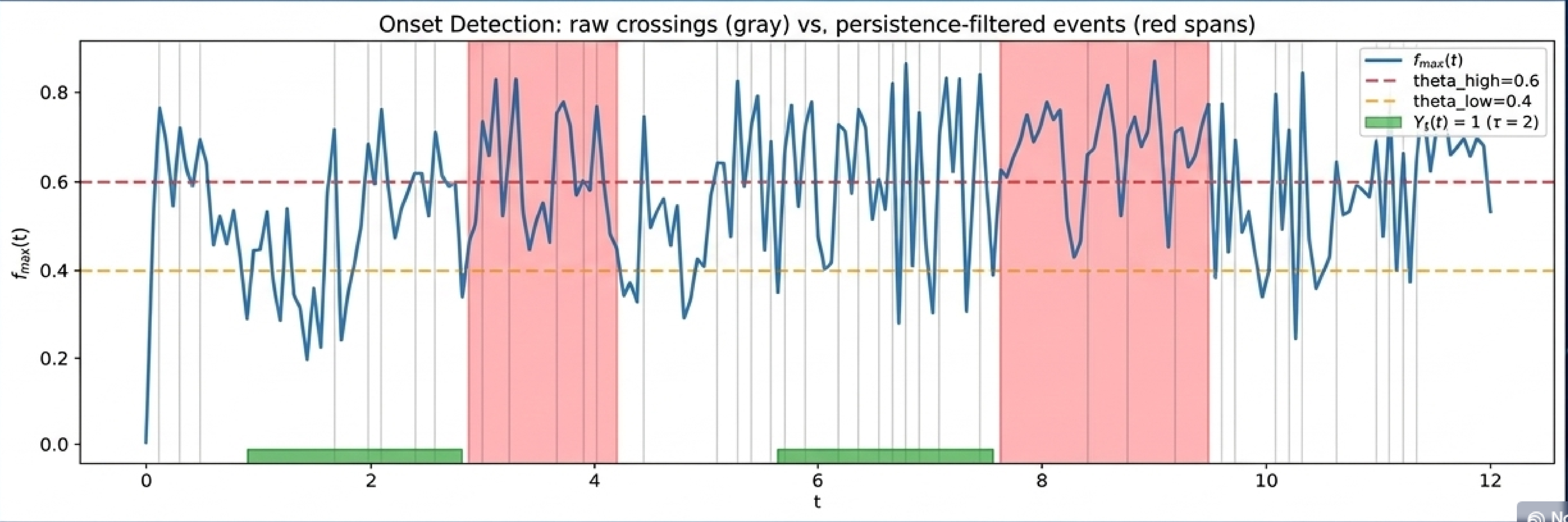
Rigor Check 1: The Measurement Artifact

The Flaw

Before running the pipeline, a validity scan caught a critical flaw: the natural order parameter (f_{max} cluster fraction) read exactly 1.000 at t=0 for dense systems. The cause was geometric, not dynamical.

The Fix

Replaced with a composite of density-domain indicators ($f_{\text{dense,max}}$, f_{void}, low-q structure factor, bimodality) evaluated via a persistence criterion rather than raw crossings.



Phase 1: Observational Identification (The Setup)

1. Raw Trajectories (Input)
2. Field Reconstruction (Density ρ_h vs. Capacity/Dissipation variants)
3. LassoCV Regression (Chronological train/test splits to prevent temporal leakage)
4. Prospective Target (Predicting $\Delta_\tau f_{\text{dense,max}}$)

Do capacity fields predict near-future density-domain structure better than contemporaneous density alone?

Observational Results: The Null Reality

$R^2 = 0.348$	$R^2 = 0.348$
Density Baseline	Density + Capacity

Across ten independent runs at Pe=130, adding any capacity or dissipation variant provided ZERO statistically significant incremental predictive value over the density-only baseline.

Capacity features behave as downstream expressions of density-driven clustering rather than as sources of independent predictive information.

Proving the Null: The Stability Battery

✓	**Replicate Count:** Null strengthened when increasing from 5 to 10 runs.
✓	**Coarse-Graining:** Held cleanly across $h=0.8$ and $h=1.8$ bandwidths.
✓	**Activity Level:** Held entirely at elevated $Pe=200$.
✓	**Packing Fraction:** A minor anomaly at $\phi=0.35$ was investigated and failed to replicate at adjacent fractions, confirming the null.

Conclusion: This is a structurally robust "Clean Failure" on the predictive front.

Phase 2: The Causal Intervention Framework

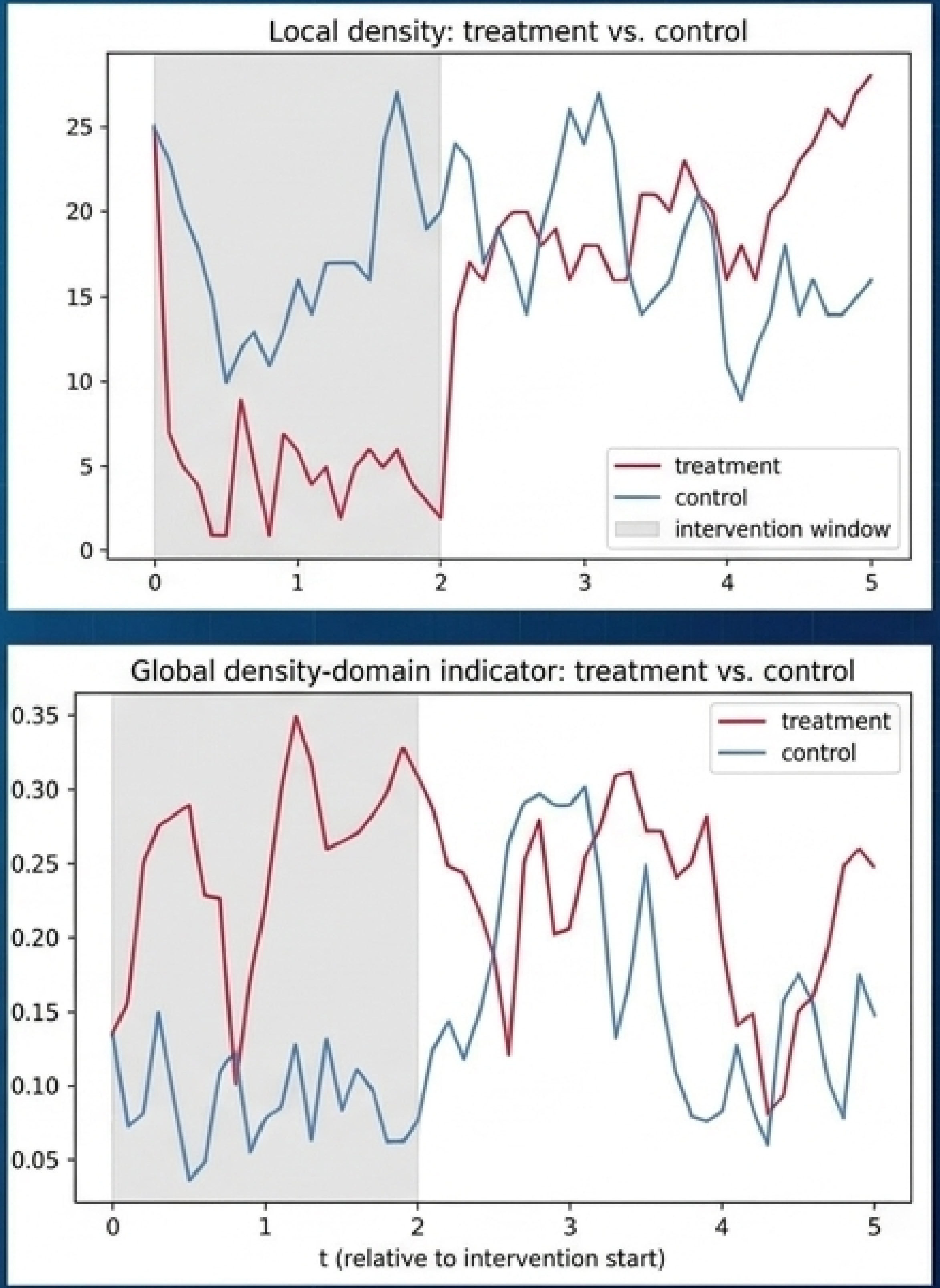
Observational prediction is not causation. To test if capacity is a driver, we built a counterfactual intervention.

The Mechanism: Cloning a state and advancing both branches with independently instantiated, identically seeded random-number generators (Common Random Numbers). At zero amplitude, the two branches are bit-identical. Any divergence is strictly due to the intervention.

Rigor Check 2: Validating the Intervention

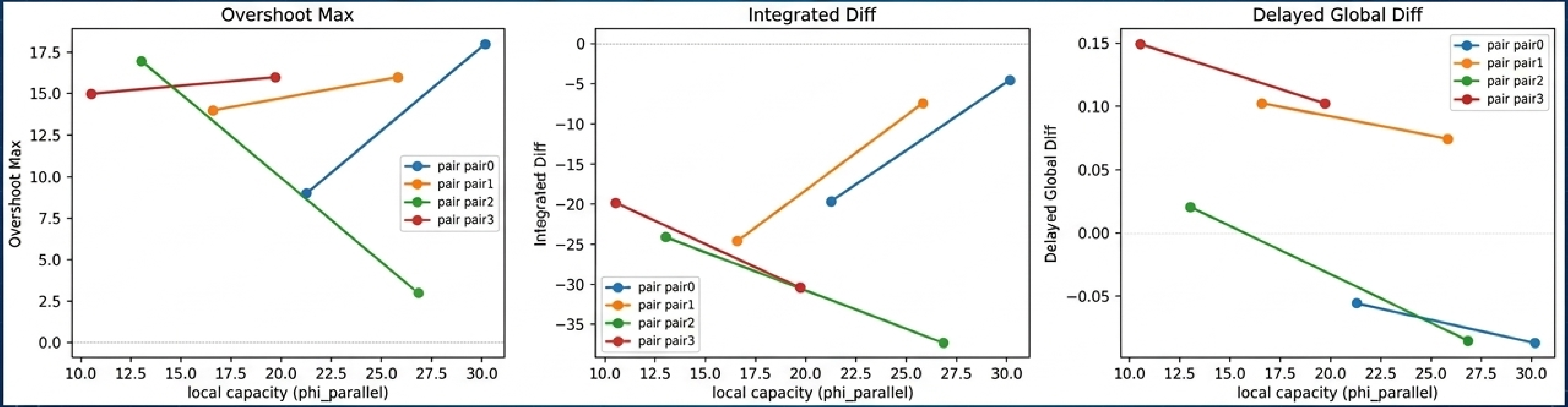
The Bug: An early timing logic evaluated against the absolute internal clock instead of elapsed experiment time, rendering the pulse inactive. A purely numerical output check could not have caught this.

The Validation: A positive-control check proved the fixed intervention causes a massive localized particle depletion during the pulse, followed by a distinct overshoot beyond the control trajectory. The causal lever works.



Causal Pilot Results: An Open Question

The intervention was applied to 4 density-matched, capacity-contrasting region pairs. The local response magnitude showed no consistent direction: 2 pairs responded stronger in high-capacity regions, 2 showed the exact opposite.



Status: This is explicitly an underpowered pilot. It does not resolve the causal question, but precisely specifies the design required for a statistically powered follow-up.

The Final Verdict

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Track 1 (Observational) lands on Clean Failure: The decomposition does not improve near-term prediction of MIPS-relevant structure beyond density.

Track 2 (Causal) lands on Indeterminate: Whether capacity identifies heterogeneous causal susceptibility to intervention remains an open, fully instrumented question.

The Value of Principled Failure

The capacity-dissipation decomposition tested here is observationally redundant with density. Yet, this methodological workflow—reconstruction, competing definitions, robustness testing, and intervention design—successfully narrowed the space of viable theoretical claims.

The appropriate response to a negative result is neither concealment nor overreach, but a precisely scoped statement of what failed and what remains untested.